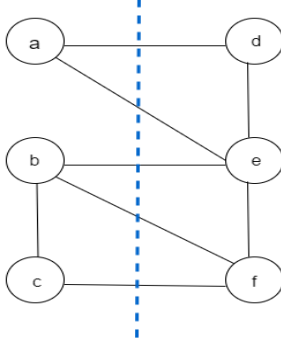


 BRAC UNIVERSITY Inspiring Excellence	BRAC University Department of Computer Science and Engineering Semester: Fall 2024 Course Code: CSE460 Course Title: VLSI Design	Final Exam Full Marks: 15 x 3 = 45 Time: 1 hour 30 minutes Date: 11th January 2025
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Set A

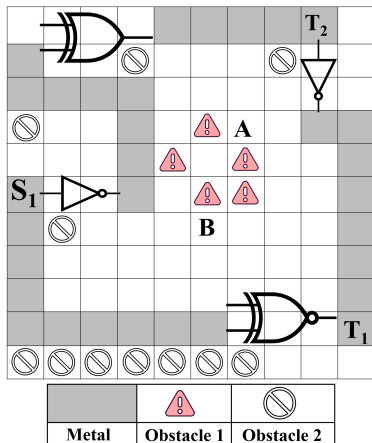
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[Answer any **THREE** questions out of **FOUR**. Each question carries **15 marks**. Return the question paper with the answer script.]

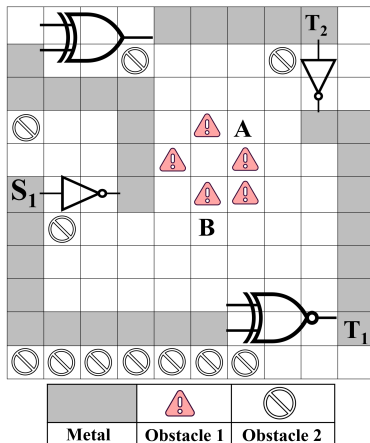
1. (CO3)	An initial cut for a partition of 6 circuit elements (a, b, c, d, e, and f) is given in Figure 1 . The cut line divides the circuit elements into two partitions.			
	(a)	Identify how many iterations are required in a single pass.		[1]
	(b)	Calculate the initial cut cost and cost of moving each node.		[4]
	(c)	Perform the first pass of the algorithm. For every “i”th iteration inside the first pass, until all the nodes are swapped and fixed, do the following: i. Compute the node costs of all unfixed nodes ii. Find the maximum gain of swapping a pair of nodes (Δg_i) iii. Swap the pair and draw the updated graph Finally, find the maximum G_m and redraw the optimized graph.		[9]
	(d)	Should you perform subsequent passes of the algorithm? Why or why not?		[1]
2. (CO2)	A facility room has 3 temperature sensors placed at 3 corners, each with two output pins: - Analog temperature value pin (not considered here). - Green LED pin that indicates whether the sensor is functional: - High (1): Sensor is functional, and the LED is on. - Low (0): Sensor is not functional, and the LED is off. If all three sensors (A, B, C) stop working simultaneously, an alarm must be triggered. Even if only one or two sensors stop working, the remaining functional sensors provide temperature data correctly. Jim and Campbell are tasked to design a logic device that: - Triggers an alarm (output = HIGH) when all three LEDs are LOW. - Does nothing (output = LOW) otherwise. Jim finally decided to implement a 3-input XOR gate, whereas Campbell chose a 3-input NOR gate to use as the alarm trigger. Take A, B, C as the inputs to the gates where A, B, C indicate the LED status (ie ON or OFF) of 1st, 2nd and 3rd sensor respectively. [Choose widths of PMOS and NMOS transistors to achieve worst-case effective rise and fall resistance equal to that of a unit inverter (R)]			
	(a)	Whose design is correct- Justify .		[1]
	(b)	Draw the CMOS circuit of the correct logic function and derive the RC equivalent circuit.		[4]
	(c)	Calculate t_{pdf} , t_{cdf} , t_{pdr} , t_{cdr} for the circuit if the output is driving 'h' number of 2-input NAND gates as its load [Choose widths of PMOS and NMOS transistors to achieve effective rise and fall resistance equal to that of a unit inverter (R)]		[4]
	(d)	Identify the parasitic and effort delay from the t_{pdf} .		[2]
	(e)	For $\lambda = 10$ nm, if $R = 5$ k Ω . μ m, $C = 20$ fF/ μ m, find the values of the delays whose expressions you found in (c). Take $h=5$.		[4]

3. (CO4)	For question (a) you are given a certain chip planning task. You have four blocks with the following widths and heights: Block A: $w = 1, h = 5$ or $w = 5, h = 1$ Block B: $w = 1, h = 3$ or $w = 3, h = 1$ Block C: $w = 2, h = 1$ or $w = 1, h = 2$ Block D: $w = 3, h = 3$ or $w = 1, h = 9$ For questions (b-e) refer to the figures at the bottom of the question. You can see the grid layout of a system with a source (S1) and two targets – T1 and T2. The obstacles and free cells are shown.	
(a)	Find and draw a floor plan with the minimum total area enclosed by the global bounding box using 10x10 space.	[3]
(b)	Determine the minimum Manhattan distance from A to B (You don't have to show in grid. Just write the answer).	[1]
(c)	Using the <u>Lee's Maze Algorithm</u> , find the shortest path from S1 to T1 and T2 <i>allowing minimum bends</i> . Answer this question on this and the subsequent grids on the bottom of the question paper. A and B are not obstacles.	[4]
(d)	Find the total memory usage of the given grid if the sequence is a series of natural numbers. If the maximum memory requirement of the provided grid is 27 bytes, calculate the maximum number of bits that each cell can hold.	[2+3]
(e)	Find the number of consecutive and unique numbers (starting from 0) that can be used for wave propagation in the Lee's Maze algorithm, for the given memory constraint in (d). What should be a full sequence for wave propagation?	[2]
4. (CO1)	A chip contains 1 billion transistors made from a 0.7 V 120 nm technology with $\lambda = 60$ nm. Out of them, 100 million are logic gates, 200 million are never on, and the rest are memory arrays. Suppose that the logic gates operate at 1 GHz, the memory arrays operate at 3 GHz, and the clock frequency is 10 GHz. For all transistors, the gate and diffusion capacitance are 1.2 fF/um and 1.7 fF/um. The average width for logic gates and memory arrays are 4λ and 5λ, respectively. The following power data are given for the entire chip, • Gate leakage power = 10 W • Junction leakage power = 20 W • Subthreshold power = 5 W • Short circuit power of the logic gates = 60 W • Short circuit power of the memory arrays = 70 W The maximum acceptable power consumption of the chip is 1 kW.	
(a)	Find the activity factor of the logic gates and the memory arrays.	[2]
(b)	Find the total switching power of each of the three kinds of transistors.	[6]
(c)	Find the static, dynamic, and total power of the entire chip.	[3]
(d)	Is the total power consumption within the acceptable range? If not, find the supply voltage that will keep it within range.	[4]

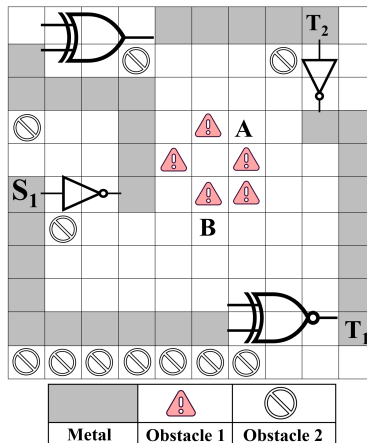
Grid 1



Grid 2



Grid 3



Grid 4

